

Basics of Innate Immunity I

Innate cells, pattern-recognition receptors, cytokines, complement, and complement regulation

If you remember nothing else

- Know the major cells that participate in innate immune responses and the roles they play.
- Pattern-recognition receptors detect recurring microbial or danger-associated patterns.
- Pro-inflammatory and anti-inflammatory cytokines coordinate the intensity and direction of innate responses.
- Complement can attack pathogens directly, opsonize them, amplify inflammation, and interact with immune cells.

Complement map

- Classical, lectin, and alternative pathways converge on shared downstream complement functions.
- Complement is powerful enough that regulation is essential; loss of regulation can produce disease.
- The lecture specifically connects complement-regulatory disorders with paroxysmal nocturnal hemoglobinuria and angioedema.

PRR families emphasized

- Toll-like receptors (TLRs).
- NOD-like receptors (NLRs).
- RIG-like receptors (RLRs), STING, C-type lectin receptors, scavenger receptors, and complement receptors.

Exam frame

- Separate the initiating pathway from the shared complement effector functions.
- When a vignette is about uncontrolled complement activity, think about a failure of regulation rather than simply 'too much immunity.'
- Keep cell type, receptor type, cytokine signal, and complement function as four separate layers.